Task 1 – Lab Document Practice & Prompting Report

Task 1 – Lab Document Practice & Prompting Report

Student Name: Danish Shahzad
Course: AI Prompting in 2026 – Lab Practice

Introduction
The purpose of this lab was to understand how modern AI systems respond to different prompting techniques and how prompt quality affects output quality. During the lab, I practiced all major concepts including context engineering, retrieval modes, reasoning prompts, multimodal inputs, brainstorming workflows, data analysis prompting, model comparison, and prompt iteration.

1. What I Learned From the Lab
• Context dramatically improves AI responses.
• Confidence does not always mean accuracy.
• Different retrieval modes (memory, search, research) produce different results.
• Structured prompts generate better outputs.
• Reasoning prompts improve decision-making quality.
• Prompt iteration is one of the most powerful techniques.
• Multimodal AI can analyze images and documents.
• AI can generate small applications through clear instructions.
• Data analysis should be verified with code execution.
• Different AI models have different strengths.

2. How My Prompting Improved
Initially, my prompts were short and vague, resulting in generic answers. After applying context, constraints, audience information, and formatting requirements, responses became more personalized, accurate, and actionable.

3. Techniques That Helped Me Most
1. Context Engineering
2. Brainstorm–Iterate Loop
3. Think Hard Prompting
4. Explicit Output Formatting
5. Asking for Sources and Citations

4. Challenges Faced and Solutions
• Generic responses → Added context and constraints.
• Overconfident answers → Requested sources and citations.
• Weak first drafts → Used iterative prompting.
• Data analysis errors → Required code execution and verification.
• Model differences → Compared outputs across multiple models.

Conclusion
This lab demonstrated that prompting is a practical skill. The quality of AI output depends heavily on the quality of instructions provided. The most valuable lessons were: context matters, iteration improves results, AI confidence is not accuracy, different tasks need different prompting strategies, and verification is essential.